ANEL HODZIC

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EDUCATION

University of Adelaide **Expected graduation 2029**

Bachelor of Engineering (Honours) in Electrical Engineering with a Bachelor of Computer science (Double Degree), currently second year

Relevant Coursework: Signals and Systems, Digital Signal Processing, Digital systems, Electronic Device and Circuits, Circuit Analysis, Engineering Modelling II, Differential Equations for Engineers, Electrical Machines

TECHNICAL SKILLS

Electronics: KiCad, LTSpice, PCB design (4-layer, SMD), soldering, multimeter, ESP32, Raspberry Pi, FPGA (SystemVerilog, Vivado)

Mechanical/CAD: Autodesk Inventor, Fusion 360, 3DEXPERIENCE, 3D printing, welding and steel fabrication

Software: C, C++, Python, MATLAB, Linux (CLI)

Automotive: Scan tool live data, fault diagnosis, electrical continuity and voltage testing

PROJECTS

Robot Vacuum: BLDC Motor Control, ESP32, Infrared Mapping

- Designed a robot vacuum with a brushless DC motor, servo positioning and infrared sensors that map a room and build a confidence map of areas to clean.
- Simulated each module in LTspice to check the voltage and current at every node, then sized PCB trace widths and heat sinks for MOSFETS and voltage regulators from the results
- Calculated voltage divider values, voltage regulator heat losses, and 555 timer PWM for the custom buck converters
- Tested the first PCB version and found faults in the second buck converter, multiplexer signals and NAND gate logic. Currently redesigning as a 4-layer SMD board with a full ground plane, digital signal lines separated from power traces, and wide power traces on the outer layer for heat dissipation.

Automotive Diagnostics: 2013 Audi A3

- Diagnosed several faulty components, including a turbo actuator fault. Used live scan tool data logged from test drives, identified the faulty component by analysing the faulty data, then replaced, calibrated and tested the new component.
- Traced dead front sensors to a damaged connector rather than sensor or wiring loom, using multimeter continuity and voltage checks. Disassembled and repaired the connector.
- Diagnosed a faulty Daytime Running Light to a water damaged module. Replaced the module and resealed the headlight housing.
- Isolated faults by testing parts individually, including the diverter valve and camshaft actuator.

Go Kart:

- Fabricated a go kart from scratch including welding steel, designing the steering system and 3D printing a racing steering wheel.
- Designed a PWM speed control PCB and calculated optimal gear ratios for balanced speed and acceleration.

Other Projects: C++ trading simulator with Black-Scholes pricing and options trading (team of 3); MATLAB program that tracks live camera motion and creates bounding boxes for the objects that move most; FPGA calculator, PWM, as well as sound signal processing projects such as creating echo's and storing sound data; Hall Effect interactive LED Chessboard that tracks moves and shows legal chess moves.

WORK EXPERIENCE:

HodzicTutoring 14th March 2025 – Present

Founder and Tutor

- Grew to 30+ clients in the first 2 years, teaching maths, physics, and chemistry from primary education to university level.
- Designed and implemented Facebook ads and business cards and set up an Instagram and Facebook page to reach new clients.

Coles, Fresh Produce (Supermarket) 10th May 2023 – Present

Sushi Train, Waiter 2022 – 2025

AWARDS

Australian Submarine Scholarship (AUKUS), 2026 and 2027

School Dux, Paralowie R-12

Silver Medal, NASSSA Science Olympics, Salisbury region, Year 12 (team)

Gold Medal, NASSSA Math Olympics, Year 10 (team)